

Jiapeng Gao

PhD Student, School of Physics, Georgia Institute of Technology, Atlanta, GA, USA

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GitHub: github.com/Gaojiapeng Website: gaojiapeng.github.io

Education

Georgia Institute of Technology, Atlanta, GA, USA

Ph.D. in Physics (expected Fall 2028). Advisor: Prof. Gongjie Li

Aug 2023 – Present

M.S. in Physics (received Aug 2025)

Aug 2023 – Aug 2025

Imperial College London, London, UK

Oct 2021 – May 2023

Graduate studies, MSc Physics with Extended Research. Advisor: Prof. Carlo Contaldi

Huazhong University of Science and Technology (HUST), Wuhan, China

Sep 2016 – Jun 2020

B.S. in Physics

Thesis: “Multi-Messenger Astronomy of Double White Dwarfs in the Milky Way”

Advisor: Prof. Yan Wang. Awarded Outstanding Undergraduate Thesis (top 3% of department)

University of California, Berkeley, CA, USA

Jun 2018 – Aug 2019

UC Berkeley Extension (Jan – May 2019) and Summer Programs (2018, 2019)

Professional Appointments

- Research Assistant, Center for Gravitational Experiments, HUST

Sep 2020 – Sep 2021

Honors and Awards

- Bayforest Technologies Limited Prize for Best Project – First Prize, Imperial College London 2023
- HUST Outstanding Undergraduate Thesis (top 3% of department) 2020
- Scholarship awarded by NAOC & Purple Mountain Observatory 2019
- Kwang-Hua Scholarship for excellence in character and learning (top 2% of department) 2017
- Group Third Prize, China Undergraduate Physics Tournament (CUPT) 2017

Publications

- **J. Gao**, G. Li, “Formation of Warm Sub-Saturns through Structure-Coupled von Zeipel–Lidov–Kozai Migration,” manuscript under review.
- C. Chen, **J. Gao**, L. Kong, G. Li, M. Tao, “DeepTTV: Deep Learning Prediction of Hidden Exoplanet From Transit Timing Variations,” submitted.
- H. Gautham Bhaskar, N. W. H. Moore, **J. Gao**, G. Li, B. Quarles, “Main-sequence systems: orbital stability around single star hosts,” *arXiv e-prints*, arXiv:2407.13899 (2024).

Presentations

- **J. Gao**, “Coupling exoplanet interior with orbital dynamics,” Oral, Exoplanet Summer Program, Other Worlds Laboratory, UC Santa Cruz, CA, Aug 20, 2026.
- **J. Gao**, Y. Tang, G. Li, J. Fortney, “Constraining the Interior Structure of GJ 436b with a Coupled Structure & Dynamical Evolution Model,” Poster, 57th Annual DDA Meeting, Chicago, IL, Jun 2026.
- **J. Gao**, G. Li, “Coupling radius inflation with high eccentricity migration: An efficient formation channel for close-in puffy planets,” Oral, AAS 245th Meeting, National Harbor, MD, Jan 2025.
- **J. Gao**, G. Li, “How to bake puffy planets: Coupling radius inflation with high eccentricity migration,” Early Career Talk, ExoPAG 31, National Harbor, MD, Jan 2025.

Research Experience

Exoplanet Formation and Dynamics

Aug 2023 – Present

Advisor: Prof. Gongjie Li, Georgia Institute of Technology

- Developed a structure-coupled von Zeipel–Lidov–Kozai (ZLK) migration model that links planetary radius inflation with high-eccentricity migration, demonstrating a formation channel for warm sub-Saturns (first-author manuscript under review).
- Applied the coupled structure and dynamical evolution model to constrain the interior structure of the warm Neptune GJ 436b (with Y. Tang and J. Fortney).
- Contributed to DeepTTV, a deep learning framework for predicting hidden exoplanets from transit timing variations.
- Contributed to a study of planetary orbital stability around single main-sequence host stars.

Constraining Lorentz Violating Parameters with EMRIs

May 2022 – 2023

Advisor: Prof. Carlo Contaldi, Imperial College London (awarded best project)

- Developed code incorporating Lorentz-violating parameters into EMRI gravitational waveforms.
- Performed a Fisher information matrix analysis forecasting a tighter constraint on the graviton mass than current bounds.

Multi-Messenger Astronomy of Double White Dwarfs in the Milky Way

Oct 2019 – Jun 2020

Advisor: Prof. Yan Wang, Center for Gravitational Experiments, HUST

Undergraduate thesis (awarded Outstanding Undergraduate Thesis)

- Improved an existing method for generating Galactic double white dwarf (DWD) population samples.
- Simulated DWD observations with LSST, CSST, LISA and TianQin, and characterized the detectable population (chirp masses, apparent magnitudes, orbital periods).
- Demonstrated that TianQin localization can guide optical follow-up, and assessed the potential of CSST for DWD searches.

Projects

ExoplanetML — Group Project Leader

Fall 2025

Group 51, ML 7641 Machine Learning, Georgia Institute of Technology

- Led the group project and built a machine learning algorithm to process exoplanet light curve data and identify false positives.

Teaching

Teaching Assistant, Georgia Institute of Technology

- PHYS 4147 Relativity Fall 2024
- PHYS 2211 Intro Physics Summer 2024
- PHYS 3021 Stellar Astrophysics Spring 2024

Other Teaching

- Teaching Assistant, Innovative Base of Physics Experiment, HUST Sep 2017 – Sep 2018
- Taught basic astronomy to students at the Middle School Attached to HUST Fall 2017

Service and Outreach

- Referee for *The Astrophysical Journal* and *The Astrophysical Journal Letters*
- President, HUST Association of Astronomy Enthusiasts Sep 2017 – Sep 2018
Club rated four-star (9th of 80+ clubs) during presidency

Public Talks

- “Earth as an Exoplanet,” Atlanta Astronomy Club, Atlanta, GA May 17, 2025
- “Earth as an Exoplanet,” Astronomy on Tap, Atlanta, GA Apr 3, 2025